Output-error sensitivity of a Clements unitary mesh vs. mean insertion loss (small range; fixed $\sigma_L = 0$ dB)

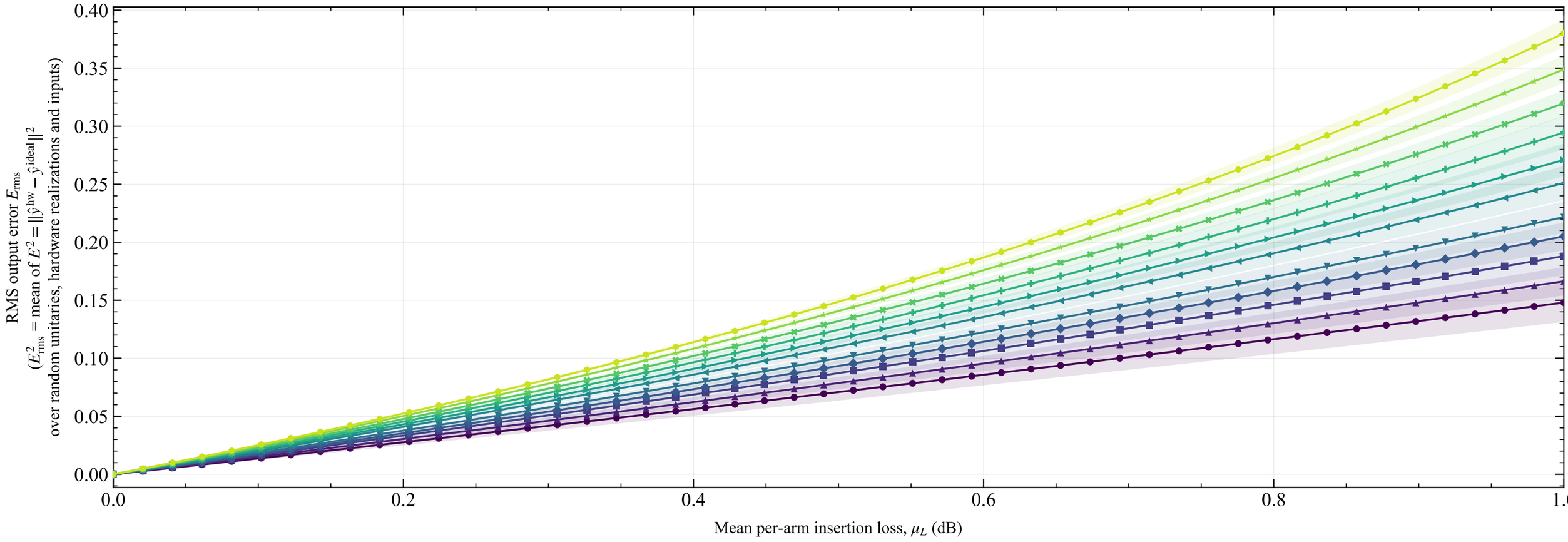
$N = 20, 28, 38, 53, 72, 100, 138, 190, 263, 362, 500$ | Haar-random unitary U (single Clements mesh) | one forward pass $y = Ux$ | $\hat{y}$ = complex output field, normalised to unit L2 norm | $E = \|\hat{y}^{\mathrm{hw}} - \hat{y}^{\mathrm{ideal}}\|_2$

Inputs per unitary: 7 types x 10 random draws (dense complex, dense real, binary ±1, phase-only, sparse 10%, single port, Gaussian beam) | shown: mean over all 7 types (equal weight)

30 hardware realizations per unitary per point (each chip sees one input of every type; each input used on 3 chips) | 25 Haar unitaries | 50 points in $\mu_L \in [0, 1]$ dB | fixed $\sigma_L = 0$ dB, all other error channels ideal

Hardware model: per-arm loss $L \sim \mathrm{TruncNorm}(\mu_L, \sigma_L^2; 0, 2\mu_L)$ dB, amplitude $10^{-L/20}$, independent for each arm of every MZI and every chip; single Clements mesh, ideal phase shifters and 50:50 couplers, ideal output phase screen

Each point: $E_{\mathrm{rms}} = \sqrt{\langle E^2 \rangle}$ over unitaries, chips and input types; band: ±1 s.d. across Haar-random targets (s.e. in the CSV)



Network size N | loss spread $\sigma_L = 0$ dB (fixed)

$N = 20$ (slope $= 0.136$ dB^{-1})	$N = 28$ (slope $= 0.1494$ dB^{-1})	$N = 38$ (slope $= 0.1628$ dB^{-1})	$N = 53$ (slope $= 0.1718$ dB^{-1})
$N = 72$ (slope $= 0.1824$ dB^{-1})	$N = 100$ (slope $= 0.1968$ dB^{-1})	$N = 138$ (slope $= 0.2063$ dB^{-1})	$N = 190$ (slope $= 0.2163$ dB^{-1})
$N = 263$ (slope $= 0.2258$ dB^{-1})	$N = 362$ (slope $= 0.237$ dB^{-1})	$N = 500$ (slope $= 0.2459$ dB^{-1})	